Varendra University

Dept. of Computer Science & Engineering (CSE)

Microprocessor, Microcontroller

Course Title: **Microprocessor and Assembly Language**
Course No: **CSE 3205**

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Outline

1. Microprocessor.
2. Microcontroller.
3. Difference between microprocessor and microcontroller.

Microprocessor

Microprocessor is a processor which incorporates the functions of a CPU on a single integrated circuit (IC).

Types of microprocessor:

1. 8086
2. 8085
3. 8088

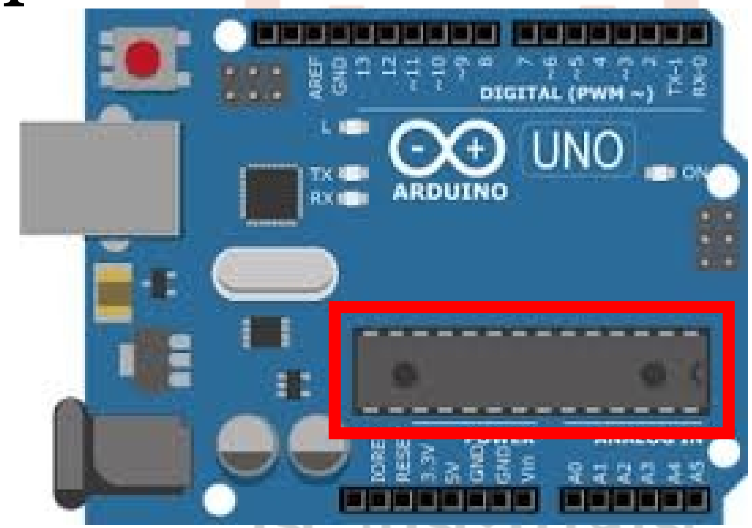
Application of microprocess

1. Personal computer.
2. Mobile device.

Microcontroller

Microcontroller is a compact integrated circuit designed to govern a specific operation in an embedded system.

Embedded systems are focused on controlling and monitoring specific functions within a device.



Application of microcontroller:

1. Washing Machine.
2. Fire detection and safety device.

Difference between microprocessor and microcontroller

Microprocessor	Microcontroller
It is used for big application.	It is used to execute a single task.
It is the heart of computer system.	It is the heart of the embedded system.
It is just a processor. Memory and I/O components have to be connected externally.	It contains external processor along with internal memory and I/O components.
Power consumption is high.	Power consumption is low.
Cost is more.	Cost is less.
Difficult to replace.	Easy to replace.
Application: Personal computer.	Application: Washing machine.